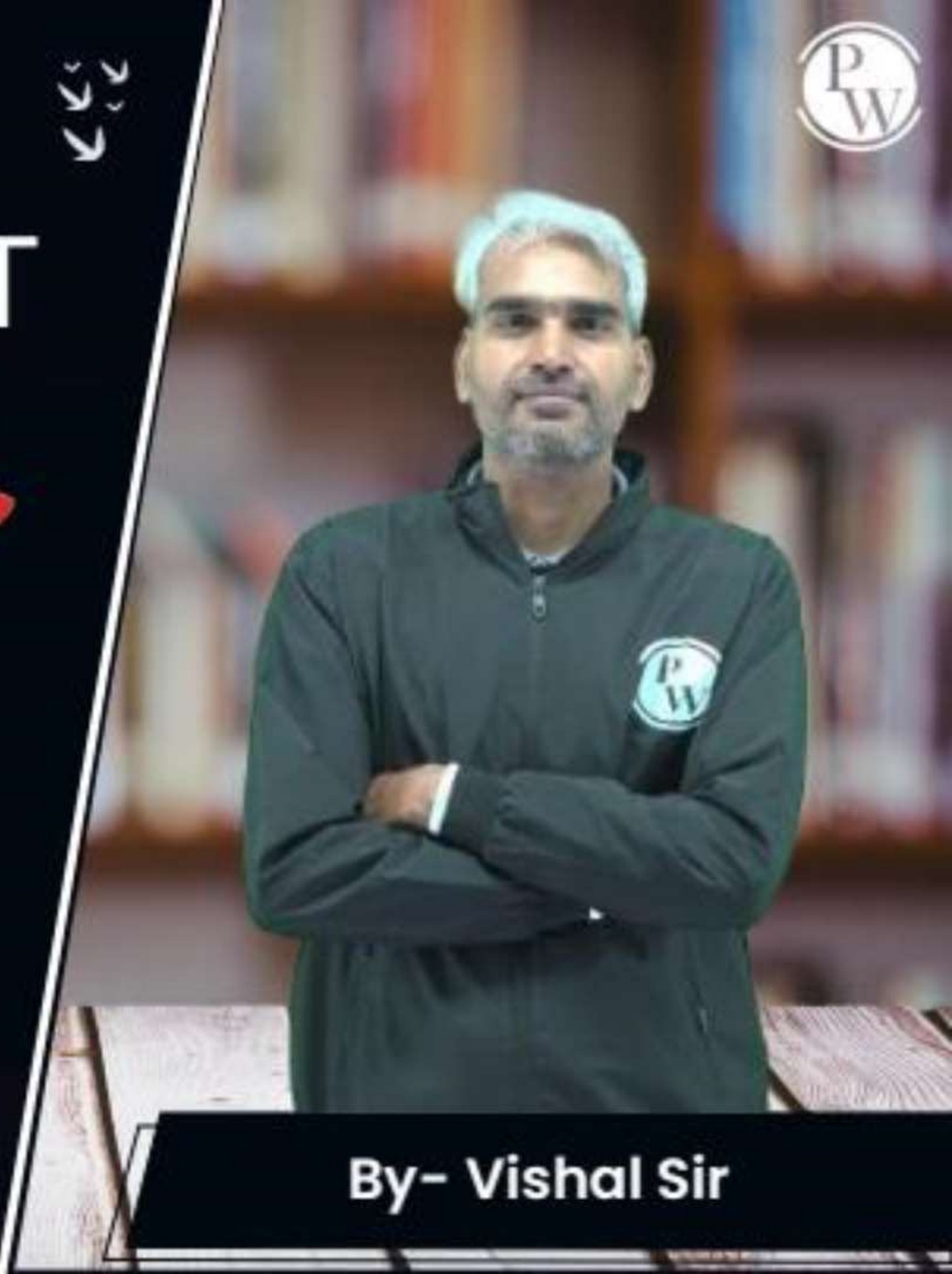
Computer Science & IT



Database management system

Transactions and concurrency control

Lecture No. 07



By- Vishal Sir

Recap of Previous Lecture



- **Topic** Conflicting and non conflicting operations
- **Topic** Conflict serializable schedule
- **Topic** Precedence graph
- **Topic** Topological ordering

Topics to be Covered



Topic

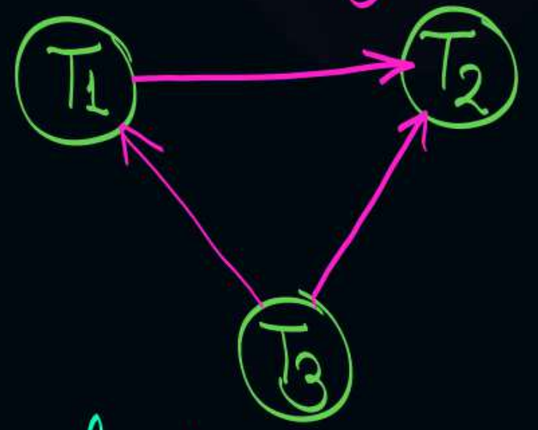
Practice questions

Slide

Q:- ⋆ Check whether the schedule is C.S.S. or not ?
If Conflict serializable schedule, then identify all Conflict equivalent serial schedules "Precedence graph"

S_1

T_1	T_2	T_3
		$R_3(y)$
		$R_3(z)$
$R_1(x)$		
$W_1(x)$		
		$W_3(y)$
		$W_3(z)$
	$R_2(z)$	
$R_1(y)$		
$W_1(y)$		
	$R_2(y)$	
	$W_2(y)$	
	$R_2(x)$	
	$W_2(x)$	

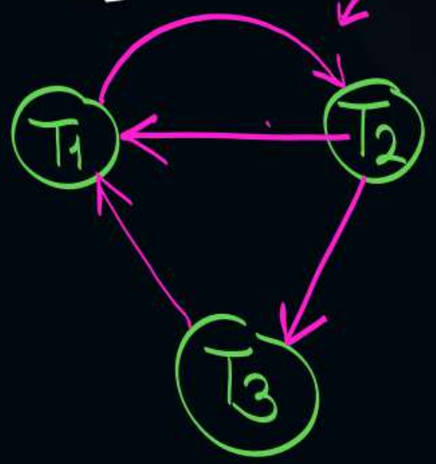


Acyclic Precedence graph
∴ "S_1" is C.S.S.
And Conflict equivalent serial schedule is $T_3 \rightarrow T_1 \rightarrow T_2$

Q:- ★ Check whether the schedule is C.S.S. or not ?
If Conflict serializable schedule, then identify all Conflict equivalent serial schedules

S_2

T_1	T_2	T_3
	$R_2(A)$	
$R_1(B)$		
	$W_2(A)$	
		$R_3(A)$
$W_1(B)$		
$W_1(A)$		
	$R_2(B)$	
	$W_2(B)$	

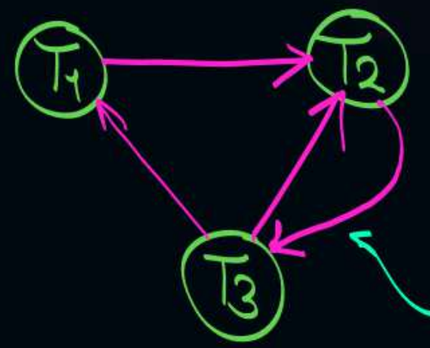


Cycle
∴ Cyclic precedence graph
Hence S_2 is not a Conflict serializable schedule

Q:- ✱ Check whether the schedule is C.S.S. or not ?
If Conflict serializable schedule, then identify all Conflict equivalent serial schedules

S_3

T_1	T_2	T_3
		$W_3(B)$
$R_1(A)$		
$W_1(B)$		
	$R_2(B)$	
	$W_2(C)$	
		$R_3(C)$

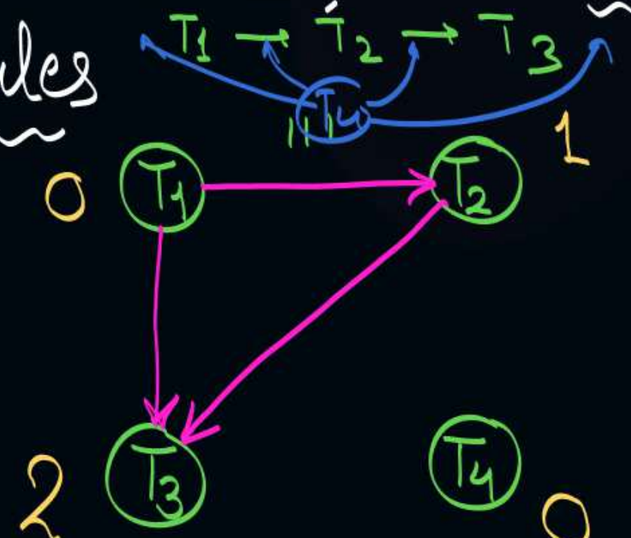


Cyclic
∴ Not a C.S.S.

Q:- ⋆ Check whether the schedule is C.S.S. or not ?
If Conflict serializable schedule then identify all Conflict equivalent serial schedules

S_4

T_1	T_2	T_3	T_4
$R_1(A)$ ✓			
	$R_2(A)$ ✓		
		$R_3(A)$ ✓	
			$R_4(A)$ ✓
$W_1(B)$ ✓			
	$W_2(B)$ ✓		
		$W_3(B)$ ✓	



Acyclic Precedence graph

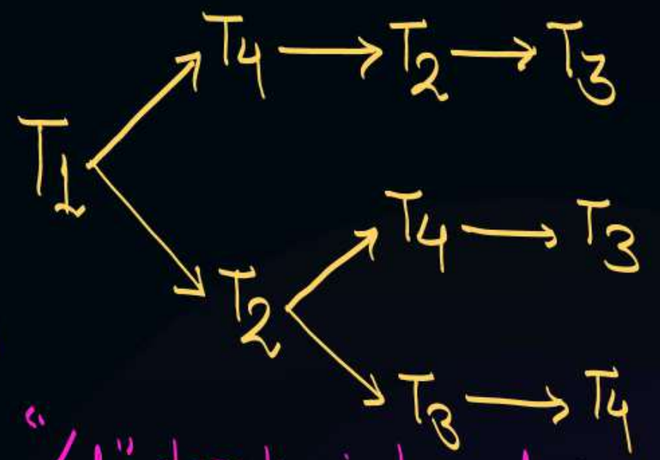
∴ Conflict serializable schedule

And Conflict equivalent Serial schedules are

$T_4 \to T_1 \to T_2 \to T_3$, $T_1 \to T_4 \to T_2 \to T_3$

$T_1 \to T_2 \to T_4 \to T_3$ and $T_1 \to T_2 \to T_3 \to T_4$

$T_4 \to T_1 \to T_2 \to T_3$



"4" topological ordering for given Precedence graph.

Note :-

(1) If schedule is C.S.S., then schedule is a Serializable schedule.

(2) If schedule is not a C.S.S., then the given schedule may or may not be Serializable

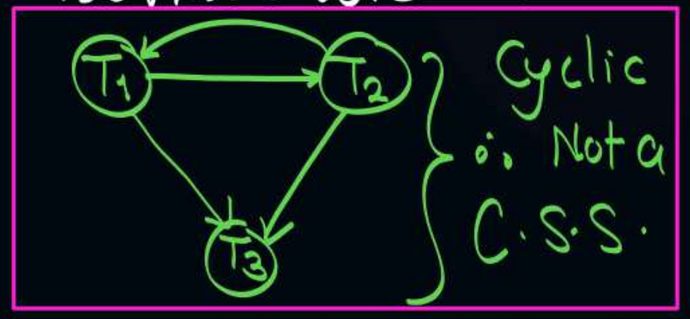
Q:- ✱ Check whether the schedule is C.S.S. or not ? If not a Conflict serializable schedule, then check whether the schedule is a serializable schedule or not

S_5

T_1	T_2	T_3
$R_1(A)$		
	$W_2(A)$	
$W_1(A)$		
		$W_3(A)$

Read from initial DB

A is finally updated by T_3



Cyclic ∴ Not a C.S.S.

Serial schedule $T_1 \rightarrow T_2 \rightarrow T_3$

Read from initial DB (same as S_5)

T_1	T_2	T_3
$R_1(A)$		
$W_1(A)$		
	$W_2(A)$	
		$W_3(A)$

A is finally updated by T_3 (Same as S_5)

$S_5 \equiv T_1 \rightarrow T_2 \rightarrow T_3$

S_5 is not a C.S.S. but its behaviour is equivalent to serial schedule $T_1 \rightarrow T_2 \rightarrow T_3$,

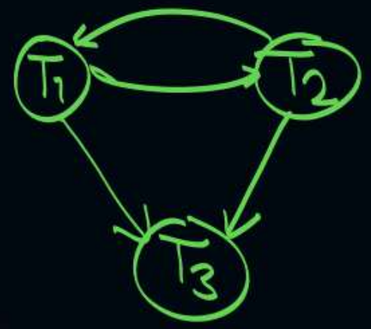
∴ S_5 is a serializable schedule

Q:- ✱ Check whether the schedule is C.S.S. or not ? If not a Conflict serializable schedule, then check whether the schedule is a serializable schedule or not

S_6

T_1	T_2	T_3
	$W_2(A)$	
$W_1(A)$		
	$W_2(A)$	
		$W_3(A)$

✱



Cyclic

∴ Not a C.S.S.

✱ Behaviour of S_6 is Equivalent to two serial schedules

$T_1 \rightarrow T_2 \rightarrow T_3$ & $T_2 \rightarrow T_1 \rightarrow T_3$

∴ S_6 is Serializable Schedule

Note:-

1. If schedule is not a C.S.S. but a serializable schedule, then at least one blind write operation will exist in that schedule
2. If schedule is not a C.S.S., and no blind write opn exists in the schedule, then schedule is a non-serializable schedule
3. If schedule is not a C.S.S, and at least one blind write opn exists in the schedule, then schedule may or may not be a serializable schedule



- Conflict serializability cond^n is, "if_then" cond^h.

i.e, if C.S.S. then Serializable

but if not a C.S.S. then we can not comment on serializability.

There are many Serializable Schedules, that are not Covered by Conflict Serializability definition

so we use "View Serializability"



2 mins Summary

Topic Practice questions

Slide

THANK - YOU

Slide